

KavachAI Investigation Report

Sovereign Industrial Agentic AI Workbench · Local Inference Verification

Investigation ID	Target Equipment	Investigation Date	Overall Status
inv_p102_det_089	Pump P-102 (Booster)	September 8, 2026	ATTENTION REQUIRED

Investigation Query:
“Investigate Pump P-102 and determine whether its condition has deteriorated.”

Overall Status: ATTENTION REQUIRED	Confidence Score: 95%	Verification Status: VERIFIED (100% Grounded)
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Key Findings & Evidence Traceability

#	Finding Description	Extracted Metric / Value	Source Evidence	Status
1	Increasing Vibration Velocity: Baseline vibration rose steadily over observed 6-month period.	Jan: 2.1 mm/s Apr: 2.8 mm/s Jul: 3.7 mm/s (+76.2%)	P-102_Inspection_Jul.pdf, p. 2 P-102_telemetry_jan_jul.csv	✓ Supported
2	Exceeds Continuous Threshold: July reading exceeds manufacturer continuous operation advisory limit.	Current: 3.7 mm/s Spec Limit: 3.0 mm/s	P-102_Pump_Operating_Manual.pdf, p. 4, §3.2	✓ Supported
3	Drive-End Bearing Temperature Rise: Temperature trend confirms friction increase.	Jan: 68°C Apr: 71°C Jul: 77°C (+13.2%)	P-102_Inspection_Jul.pdf, p. 3 ds_4471 (Telemetry)	✓ Supported
4	Topological Downstream Risk: Pump P-102 feeds directly into Hydrocracker Reactor via Control Valve.	Path: T-101 → P-102 → V-204 → R-101	pid_unit_101.png Unit_101_Interconnect_Description.pdf	✓ Supported

Deterministic Telemetry Progression (Data Agent Log)

Date / Milestone	Vibration (RMS mm/s)	Bearing Temp (°C)	Advisory Threshold	Condition Status
January 10, 2026 (Baseline)	2.1 mm/s	68.0 °C	3.0 mm/s	NORMAL
April 11, 2026 (Inspection 2)	2.8 mm/s	71.4 °C	3.0 mm/s	ELEVATED
July 09, 2026 (Inspection 3)	3.7 mm/s (+76%)	77.2 °C (+13%)	3.0 mm/s	EXCEEDED LIMIT

Conclusion & Engineering Action Plan

Synthesized Finding:

The available multimodal evidence indicates a clear condition deterioration on Booster Pump P-102 over the observed 6-month period. Vibration velocity has increased from a 2.1 mm/s baseline to 3.7 mm/s (+76.2%), crossing the 3.0 mm/s continuous operation threshold specified in the Operating Manual. Concurrently, drive-end bearing temperature has climbed from 68°C to 77°C.

Recommendation & Next Steps:

1. Schedule laser shaft alignment and coupling inspection for Pump P-102 during the upcoming maintenance window.
2. Conduct high-frequency spectral vibration analysis to isolate bearing outer-race frequency vs. unbalance.
3. Perform ultrasonic lubrication check on drive-end bearings.
4. *Note: The evidence indicates condition degradation requiring engineering attention, but does NOT establish imminent catastrophic failure.*

Automated Verification: PASSED (Confidence: 95%)
Agents Involved: Planner, Document, Data, Vision, Verification
Local Inference Host: 127.0.0.1:11434 (Qwen 2.5 3B)

Audit Checksum: sha256:7f4c029...e957
Data Sovereignty: Air-Gapped / Zero External Egress
Sign-off: KavachAI Sovereign Verification Engine